Cover Letter

Dear Topic Editors,

We would like to submit the enclosed manuscript entitled “Artificial Intelligence Readiness and Youth Employment in the European Union: Multivariate Modeling and Cluster Typologies of Youth Disengagement and Unemployment” for consideration in Systems, as a contribution to the Topic “Recent Applications of Artificial Intelligence in Economy and Society”. The manuscript has not been published elsewhere and is not under consideration by any other journal. No conflict of interest exists in its submission, and it has been approved for publication by all authors.

The topic could hardly be more timely. Artificial intelligence is most capable at precisely the entry-level analytical and administrative tasks from which early careers have always been built, and Europe monitors its young people through two headline indicators — the NEET rate and the youth unemployment rate — without knowing which of them, if either, moves with the diffusion of the technology. Using harmonised Eurostat data for the 27 member states, the manuscript measures national AI readiness on four dimensions (enterprise adoption, generative AI use among young individuals, youth digital skills, and ICT specialist employment) and relates them to both youth indicators through multivariate general linear models, an exploratory factor analysis, and a cluster typology — the comparative, systems-oriented register in which the Topic’s recent articles are written.

The highlights of the paper are as follows.

(1) A clean and policy-relevant asymmetry. All four readiness dimensions are significantly and negatively associated with the youth NEET rate (partial eta squared between 0.515 and 0.626), while none is significantly associated with the youth unemployment rate. The statistical relationship between AI and youth labour markets runs through engagement, not through measured unemployment — a distinction the literature has not previously drawn at this level.

(2) A latent readiness dimension. Principal Axis Factoring (KMO = 0.778) condenses the four indicators into a single factor explaining 76.9% of their variance, with generative AI use among the young cohort as its purest expression (loading 0.975). One standard deviation of this factor is associated with a NEET rate lower by 2.49 percentage points ($R^2 = 0.643$).

(3) A two-regime map of Europe. Hierarchical and k-means clustering converge on a two-group structure, agreeing for 25 of the 27 countries; the k-means solution comprises a higher-readiness group of 13 economies (average NEET 8.1%) and a lower-readiness group of 14 economies (average NEET 12.2%), with France’s below-average enterprise adoption placing it in the second group. The typology separates engaged from disengaged youth labour markets — not low-unemployment from high-unemployment ones — and motivates differentiated policy for the two regimes.

(4) Disciplined, transparent inference. All conclusions are presented as exploratory cross-sectional associations: diagnostics (Shapiro–Wilk, Cook’s distance), an influential-country re-estimation, and supplementary correlations with tertiary attainment, income, and R&D intensity are reported, and the insignificant unemployment-margin results are stated as such rather than explained away.

We believe the manuscript fits the Topic precisely: it addresses the call’s interest in AI’s influence on workforce dynamics, employment trends, and labor markets with comparative European evidence, methodological transparency, and directly actionable implications — applied AI skills inside Youth Guarantee offers, protected learning content in AI-augmented entry-level jobs, and infrastructure-first strategies for the lower-readiness cluster. We hope the manuscript proves suitable for the Topic, and we would be pleased to revise it in light of the reviewers’ comments.

Yours sincerely,

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